



CIS 105: Computer Applications and Information Technology Lab

School of Data Science and Information Technology, NIT

Fall'26

Lab Manual 2

Excel Lesson 1: Working with Real Data

Name: _____

Roll Number: _____

Total Tasks: 5**Total Questions: 15**

In this lab you work on 1,000 online orders. The tasks tell you what to produce, not how to click. The methods are on the Lecture 2 slides, so try each task first and open the slides only if you get stuck. Do the tasks in order and save after each one.

Dataset

Download **Lab2-dataset.xlsx** from Canvas. The sheet **Orders** has one row per order:

Column	What it contains
Order ID	Unique code of the order (ORD-10001 to ORD-11000)
Order Date	Date the order was placed (July 2025 to June 2026)
Customer Name, City, Province	Who placed the order and where
Category, Product	What was bought
Quantity, Unit Price (Rs)	How many items, and the price of one item
Discount	Discount on the order, stored as a decimal (0.15 means 15%)
Payment Method	Cash on Delivery, Card, Bank Transfer or Mobile Wallet
Status	Delivered, Pending, Returned or Cancelled
Delivery Days, Rating	Days taken to deliver, and the customer's rating from 1 to 5

Recording your answers

Save a copy of the dataset as `RollNo_Lab2.xlsx`. Add a sheet named **Answers** and write the question number in column A (Q1, Q2, ...) and your answer in column B. *Clear one filter before you start the next question.*

Task 1: Explore the Dataset

Without scrolling through the sheet, find the following and record them on the Answers sheet.

- Q1.** The Order ID of the last order in the sheet.
- Q2.** The row number of that last order.
- Q3.** The product written in cell G500.
- Q4.** The total (Sum) of the Quantity column, as shown in the status bar.

Task 2: Freeze and Format

Turn the raw sheet into a readable one.

- a. Freeze the top row so the headers stay visible when you scroll.
- b. Format the header row (A1:N1): bold, dark fill color, white font, centered, wrapped.
- c. AutoFit the width of all columns.
- d. Show the **Order Date** column as dd-mmm-yyyy (for example 01-Jul-2025).

- e. Show **Unit Price (Rs)** with a thousands separator and no decimals (for example 10,440).
- f. Show **Discount** as a percentage (for example 15%).

Q5. Change the format of cell B2 to General and record the number it shows. Then undo the change so the date is back.

Task 3: Conditional Formatting

Create these three rules. Apply them to the data rows only, so the header stays out of every rule.

- a. Every **Status** cell that contains the word Returned: Light Red Fill with Dark Red Text.
- b. Every **Unit Price (Rs)** greater than 100000: Green Fill with Dark Green Text.
- c. **Delivery Days**: Data Bars in a color of your choice.

Task 4: Sorting

- a. Sort the whole table by **Unit Price (Rs)**, largest to smallest. Record the Order ID and the City of the most expensive item as **Q6**.
- b. Sort the whole table on three levels: **City** (A to Z), then **Quantity** (largest to smallest), then **Unit Price (Rs)** (largest to smallest). Record the Order ID in the first row of data (row 2) as **Q7**.
- c. Bring the table back to its original order, so that row 2 shows ORD-10001 again.

Task 5: Filtering

Turn on the filter and answer each question. Clear the filter before every new question. Excel shows the number of matches at the bottom left of the window (**x of 1000 records found**).

Q8. How many orders are **Electronics** orders from **Karachi**?

Q9. How many orders have Status **Returned** and Payment Method **Cash on Delivery**?

Q10. How many orders have a Quantity of **10 or more**? Also write the one Category they all belong to.

Q11. How many orders were placed in **December 2025**?

Q12. Use the red color from Task 3 to filter the Status column. How many orders were **Returned** in total?

Q13. How many orders came from the province **Sindh** and were paid by **Mobile Wallet**?

Q14. How many orders have a **Rating of 1** and **8 or more Delivery Days**?

Q15. How many orders from **Lahore** have Status **Delivered** and a **Rating of 5**?

When you are done, clear all filters so that all 1000 orders show again, and save the file.

Submission

Save your file as `RollNo_Lab2.xlsx` with the Answers sheet filled in. Keep it with you, because tasks like these can come in the first long quiz and its viva.

_____ >< _____
_____ >< _____